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MULTISCALE SCORING MODEL FOR ENHANCED URBAN PERCEPTION EVALUATION

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ABSTRACT

Effective urban management, renewal, and development rely on identifying low-quality areas within the city. However, previous studies have been limited by low-volume handcraft surveys and a dearth of data sources, making it difficult to understand human perception within the urban environment. In this paper, we propose a powerful yet simple scoring model to perform street view image recognition and evaluation which utilizes both global information and feature-level semantic information of street elements, resulting in a high-precision perception model on 6 indexes (Beautiful, Lively, Safe, Wealthy, Boring, and Depressing) from Place Pulse 2.0 dataset. The model is then independently applied to a large-scale and fine-grained evaluation task of 4,384 street view images in Shameen Region, Guangzhou, providing valuable perception details and decision-making support for urban planning for the local government. We believe our work will accelerate the digitization and intelligent transformation of municipal engineering.

Index Terms— Urban perception, Multiscale integration, Built environment, Street view images, Landscape evaluation

1. INTRODUCTION

Ensuring a comfortable built environment is crucial for promoting stable societal development and enhancing residents' quality of life [1]. The ability to gather visual information about physical surroundings and the experiences that shape observers' perceptions has long been a subject of interest across various disciplines [2].

In the era of big data, the surge in the availability of street view images (SVIs) from intelligent location services platforms like Google and Baidu Map, combined with advancements in deep learning and enhancements in computer performance and computing power have rapidly made SVIs, which can accurately reflect the physical forms and attributes of geographic spaces, a crucial data source for assessing the visual qualities of large-scale built environments [3, 4, 5]. A range of machine learning and deep learning techniques have been employed to tackle human perception evaluation tasks [6, 7, 8]. A pivotal moment occurred when Dubey et al. introduced the Place Pulse 2.0 dataset, along with the Streetscore-Convolutional Neural Network (SS-CNN) framework, designed to forecast human pairwise image perceptions [9]. Building upon this dataset, researchers subsequently training image classification models [10, 11, 12] directly from image-level information. However, none of these methods considered the impact of various elements in SVIs on the overall perception results, nor did they apply their model in a real-world urban planning scenario.

In this paper, we propose a multiscale perception method that uses ViT [13] to obtain global information of the image, uses Segformer [8] to obtain pixel-level semantic information of the image, and combines them with Random Forest for evaluating urban street views. Experiments show that our method exhibits a high level of predictive accuracy on the Place Pulse 2.0 dataset, efficiently learning and capturing perceptual features within the images. We also conduct an empirical validation in the Shameen region, the results underscores the built environment distribution of the region, indicating our method as a valuable tool for urban planning and city updating.

2. METHOD

The model's architecture is depicted in Figure 1a. Initially, we undertake the normalization and classification of the image scores within the Place Pulse 2.0 dataset, denoted as $I = \{i_1, i_2, \ldots, i_n\}$. We reshape the image $i \in \mathbb{R}^{H\times W\times C}$ into a sequence of flattened 2D patches $i^p \in \mathbb{R}^{N\times(P^2\cdot C)}$, where (H, W) is the resolution of the original image, C is the number of channels, (P, P) is the resolution of each image patch, and $N = \frac{HW}{P^2}$ is the resulting number of patches. Then each image is categorized into ten discrete classes.

$$z_n^0 = [\text{class}; i_n^1; i_n^2; \ldots; i_n^N] + E_{\text{pos}}, E_{\text{pos}} \in \mathbb{R}^{(N+1)\times D} \quad (1)$$

$$z_n'^l = \text{MSA}(\text{LN}(z^{l-1})) + z^{l-1}, \quad l = 1, 2, ..., L \quad (2)$$

$$c_n = f_{\text{class}}(z_n'^l) \quad (3)$$

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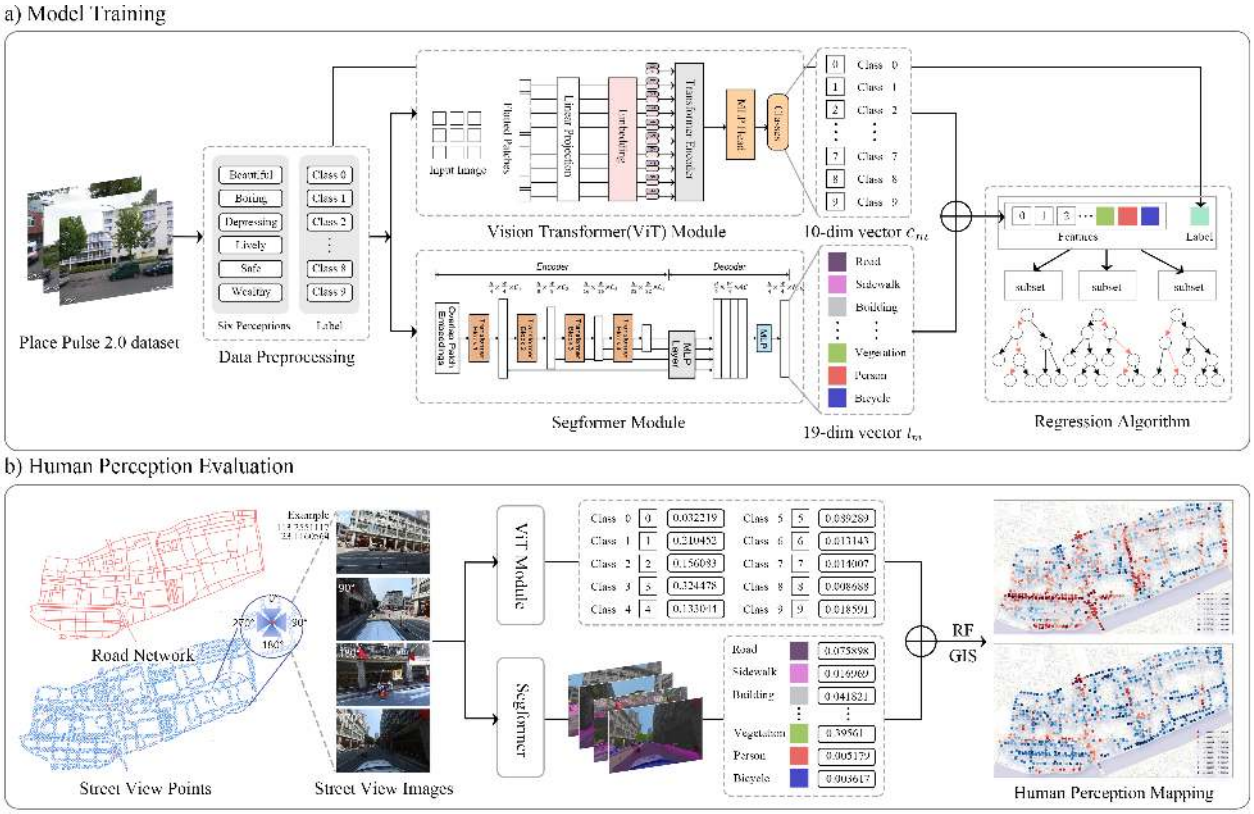


Fig. 1. a) The training process of our street view image evaluation model on the Place Pulse 2.0 dataset. b) We apply our model empirically to a external OSM-Shameen dataset, and provide the human perception mapping for future urban planning and management of the region.

The output is a set of discrete classifications $C = \{c_1, c_2, \ldots, c_n\}$, where each $c_n \in \{1, 2, \ldots, k\}$ represents the class assignment of image i_n. MSA represents L layers of multihead self-attention blocks, and f_{class} is a 10-class multi-layer perceptron (MLP) module. LN is layer normalization. The classification information is then employed to fine-tune the classification model using cross-entropy loss.

Subsequently, we use a Segformer-B5 [8] based semantic segmentation model pretrained on the Cityscapes dataset to perform pixel-level semantic segmentation. This process extracts the pixel-wise ratios $L = \{l_1, l_2, \ldots, l_n\}, l \in \mathbb{R}^{19}$ for each of the 19 object categories present in the image i_n.

$$s_n^0 = \text{CNN}(i_n) \tag{4}$$

$$s_n'^l = f_{\text{seg}}(\text{MSA}(\text{LN}(s^{l-1}))), \quad l = 1, 2, \ldots, L \tag{5}$$

$$l_n = \left[\frac{0 \text{ in } f_{\text{seg}}(s_n'^l)}{\text{total pixels}}, \frac{1 \text{ in } f_{\text{seg}}(s_n'^l)}{\text{total pixels}}, \ldots, \frac{18 \text{ in } f_{\text{seg}}(s_n'^l)}{\text{total pixels}} \right] \tag{6}$$

where f_{seg} is a multi-layer perceptron (MLP) with 19 classes, and each l_n is a pixel-wise label map for i_n. CNN stands for Convolutional Neural Network. Each label map l_n contains pixel-wise ratios for each of the object categories in the SVIs.

When evaluating on each street view image in the external OSM-Shameen dataset $O = \{o_1, o_2, ..., o_m\}$ (Fig.1b), our approach employs a two-step process. First we use the trained ViT model to encode the patchified image $o_m^p \in \mathbb{R}^{N \times (P^2 \cdot C)}$ and return the image perceptual class c_m as global information. Then, we utilize the Segformer-B5 model f_{seg} to perform semantic segmentation, generating a pixel-level semantic label l_m for street view elements. This multimodal integration technologies are widely used in other scientific problems [14, 15, 16].

The global classification vector c_m and the feature-level semantic vector l_m are then concatenated and provided as multiscale input to a Random Forest regression model.

$$p_m = \text{Random Forest}([c_m; l_m]) \tag{7}$$

where p_m is a concrete number ranging from 0 to 10, this number represents the predicted human perceptual score for the given image. The Random Forest model has learned to map the concatenated feature vector to these scores based on the training data, effectively combining the image classification outcomes and the fine-grained semantic features.

During the training phase, we train perception models independently for each human perception index. However, dur-

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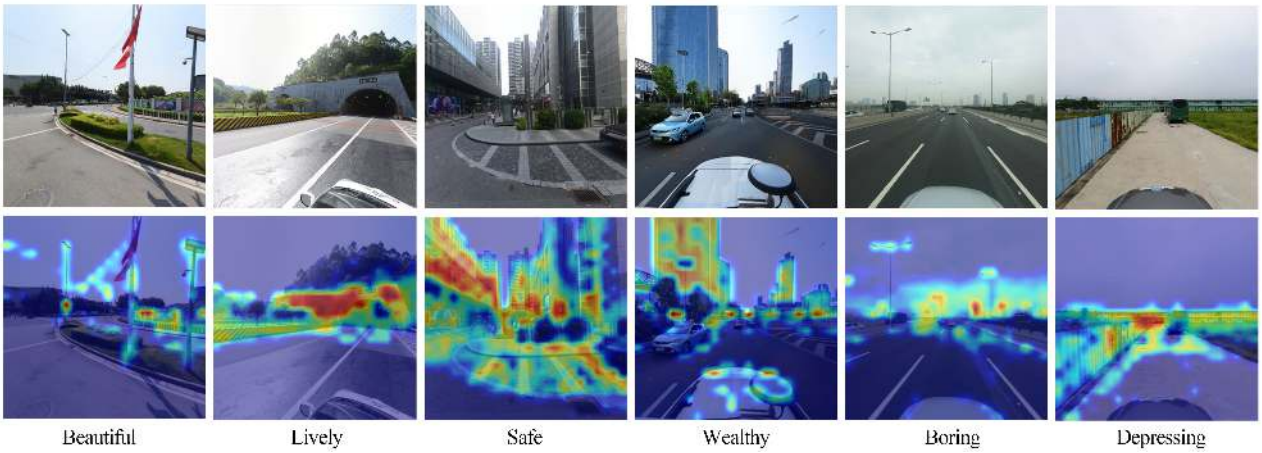


Fig. 2. Grad-CAM visualization of each of the six human perception indexes (Beautiful, Lively, Safe, Wealthy, Boring, and Depressing). Figures are randomly selected from the OSM-Shameen dataset.

Table 1. Performance Comparison of Different Models on MAE, MSE, and R^2 Metrics on six human perception indexes.

Index	Beautiful			Lively			Safe			Wealthy			Boring			Depressing		
Model	MAE	MSE	R^2	MAE	MSE	R^2	MAE	MSE	R^2	MAE	MSE	R^2	MAE	MSE	R^2	MAE	MSE	R^2
ViT	0.851	0.937	0.771	0.882	0.724	0.798	0.933	1.203	0.476	0.992	2.013	0.271	0.862	1.906	0.461	0.922	2.311	0.448
DeeplabV3 + RF	1.845	5.022	0.196	1.614	3.668	0.364	1.712	5.117	0.203	1.620	1.458	0.207	1.993	3.847	0.332	1.552	2.937	0.402
DeeplabV3 + ViT + RF	0.308	0.377	0.827	0.367	0.413	0.822	0.874	2.002	0.565	0.742	2.576	0.315	0.788	1.658	0.542	0.703	1.325	0.554
FCN + RF	1.624	4.126	0.339	1.762	3.902	0.217	1.571	4.298	0.236	1.791	3.102	0.129	2.037	5.149	0.182	1.474	2.826	0.247
FCN + ViT + RF	0.291	0.346	0.912	0.380	0.455	0.765	0.805	3.261	0.352	0.771	1.581	0.551	0.731	2.275	0.389	0.662	1.993	0.327
Segformer + RF	1.533	3.782	0.395	1.333	2.619	0.421	1.260	2.277	0.151	1.352	2.576	0.083	1.607	3.645	0.132	1.258	2.274	0.217
Segformer + ViT + RF	**0.267**	**0.335**	**0.946**	**0.239**	**0.307**	**0.927**	**0.608**	**0.893**	**0.635**	**0.532**	**1.163**	**0.669**	**0.550**	**1.210**	**0.666**	**0.550**	**1.210**	**0.666**

ing application, we ensemble these models for practical use.

3. EXPERIMENT

Datasets and preprocess. In this paper, we use two distinct datasets, one for training and one for empirical application.

Place Pulse 2.0 dataset includes 110,988 images of 1.17 million pairs from 56 cities in 28 countries across six continents [9]. These images were rated by 81,630 online volunteers based on six perceptual dimensions: safety, liveliness, boredom, wealth, depression, and beauty. We utilize the Microsoft TrueSkill algorithm [17] to obtain each image's perception score by statisticaling the results of two-by-two comparisons (considering images that have undergone comparison five times or more) [12]. Then we normalization these perception scores within a range of 0 to 10 using max-min normalization and further categorize the scores into ten classes, ranging from 0 to 9 to facilitate the forthcoming training of image classification and regression models.

OSM-Shameen dataset. We selected Shameen Region as the case study area (Fig.1b left) to exemplify the real-world application of this approach to a specific built environment. This area is situated within the historical heart of Guangzhou, covering an area of approximately 2.5 square kilometers, serving as a pivotal hub for provincial and Hong Kong water transportation, and playing a crucial role in facilitating economic and cultural exchange between China and the West. Road network of the area is first obtained through OpenStreetMap (OSM), and then we constructed street view points along the road at intervals of 25m in ArcMap 10.6. resulting in a total of 1,096 collection points. Each collection point contains four images in four directions, totaling 4,384 images, with a size of 512 x 512 pixels per image.

Experiment settings. We chose to use pre-trained ViT and Segformer models and adopted their standard hyperparameter settings to build models for each human perception index in the Place Pulse 2.0 dataset. We selected Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 metrics to assess the results of our models with human-annotated scores. We divided each dataset by 7:2:1 ratio and report the model's performance on the test dataset.

4. RESULTS

Comparison with SOTA. We conducted a thorough comparison and ablation of our model with ViT and two other

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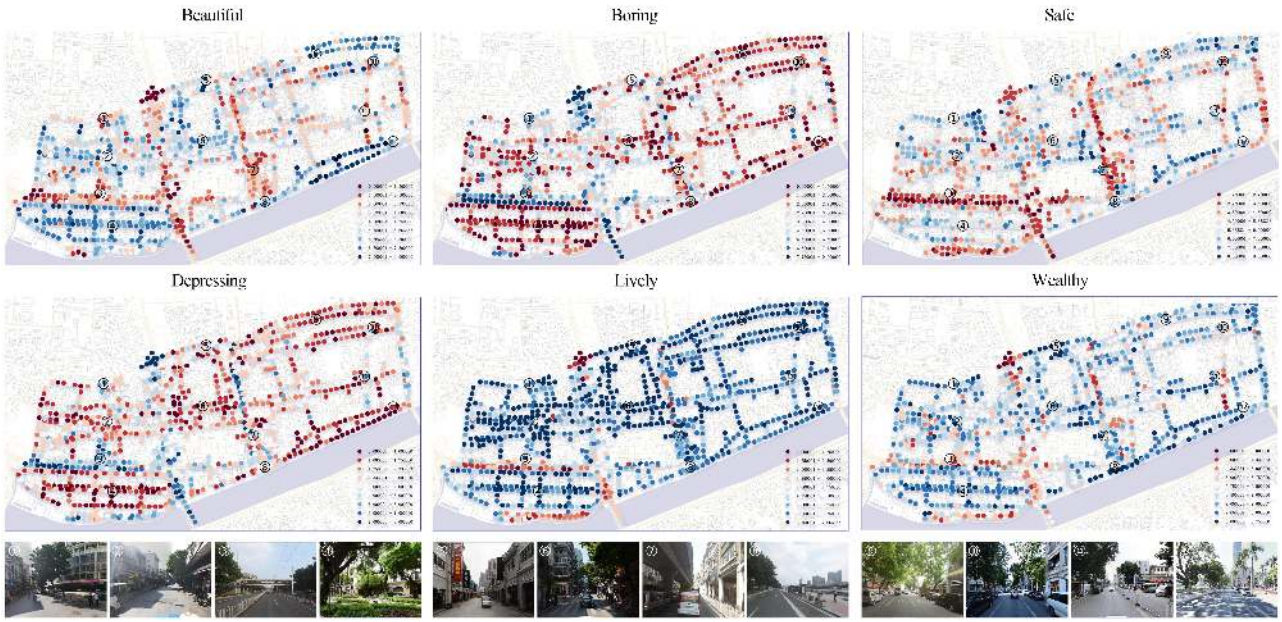


Fig. 3. Visualization of street view evaluation results on the OSM-Shameen Dataset using Geographic Information Systems (GIS). Blue points indicate high human perception scores, whereas red points indicate low scores.

SOTA models - DeeplabV3 [18] and FCN [19], and exclude the ViT classification module in each model for ablation study. We presented the predict results in Table1. In the predict results, Segformer shows better performance compared with DeeplabV3 and FCN ($\sim$ 19% MAE improvement on average compared with the second highest model), and the combination of Segformer and ViT gets the best performance under all metrics in six human perception indexes ($\sim$ 24% MAE improvement on average compared with the second highest model). The results demonstrated that our model outperformed the others in accurately capturing the human perception of SVIs. Furthermore, we observed that the addition of the ViT module substantially enhanced the model's comprehension of global information in images, leading to a significant improvement in prediction performance.

Visualization of model attention. We applied the Grad-CAM method to visualize the model's attention on the input images. We randomly select one street view image for each human perception index and draw the Grad-CAM heatmap of the image, the results are illustrated in Fig.2. As can be seen in the figure, our model can correctly attend to the index-related regions in the image. For instance, in the Wealth Index figure, the model assigns significant importance to skyscrapers, aligning with the human perception of wealth.

Application in urban planning. To apply our method in real-world scenario, we further empirically apply our street view image evaluation model to the OSM-Shameen dataset. We utilize Geographic Information Systems (GIS) to visualize the evaluation results of each human perception index at every street view points. The results are shown in Fig.3. Each figure in Fig.3 represents a human perception index.

In general, the overall vitality and prosperity of the region are relatively high, except for a few waterfront roads that score lower. Shameen island, in particular, demonstrates a distinct spatial clustering pattern, with higher positive perceptions and lower negative perceptions. This is attributed to the lush vegetation, excellent walkability, clean streets, and high spatial quality, rendering it well-suited for recreational activities and creating a pleasant environment for visitors. Such a phenomenon aligns well with the greenery level and built environment across different streets, thus validating the effectiveness of our proposed approach.

The example in Shameen Region is just one illustration of our model's application in the urban perception using SVIs. Particularly, area with high negative perception scores calls for a demand for city update, which has been taken as a reference to the government of Guangzhou.

5. CONCLUSION

In this paper, we propose a new multiscale scoring model for street view perception which integrate both global information and feature-level semantic information. Our results in Place Pulse 2.0 dataset demonstrate the high prediction precision of the human perception scores. And our empirical validation in the Shameen Region further illustrates the effectiveness of the proposed method as a decision-making support for urban management and construction. In future, we would like to extend the proposed method to include real-time updation to dynamically monitor the urban landscape environment.

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